

جامعة سطيف 1 - فرحات عباس



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MACHINE LEARNING-ASSISTED
FTIR SPECTROSCOPY
FOR CANCER BIOMARKER ANALYSIS:
FOCUS ON EGFR | ALK AND ROS1

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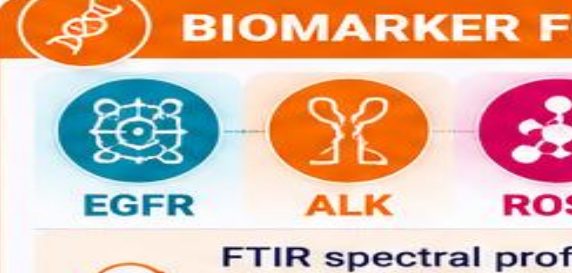
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ABSTRACT

BACKGROUND

Cancer remains a major global public health challenge and a leading cause of death worldwide. Early diagnosis significantly improves patient outcomes. Fourier transform infrared (FTIR) spectroscopy is a rapid, non-destructive, and cost-effective analytical tool for detecting cancer-related biochemical changes through molecular vibrational signatures.

BIOMARKER FOCUS



This work focuses on three clinically relevant cancer biomarkers: epidermal growth factor receptor (EGFR), anaplastic lymphoma kinase (ALK), and ROS1 proto-oncogene.

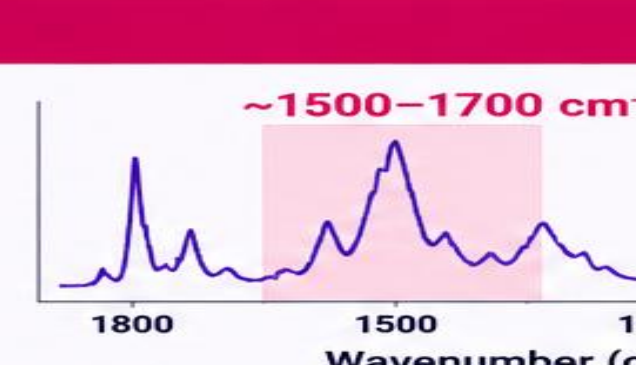
FTIR spectral profiles reflect alterations in proteins, lipids, nucleic acids, and carbohydrates, including amide I, amide II, phosphate, and lipid-related absorption bands.

METHODS



Dataset: 1,000 FTIR spectra
Four balanced classes: Healthy, EGFR, ALK, ROS1 (250 spectra per class)
Models: Logistic Regression, Random Forest, XGBoost
Analyses: PCA, ROC, SHAP, Cross-validation

RESULTS



XGBoost achieved the highest test accuracy: **97.50%**;
Random Forest achieved the highest multiclass ROC-AUC: **0.9983**.
XGBoost class-specific ROC-AUC values ranged from **0.9961 to 1.0000**.
SHAP analysis identified the most influential wavenumbers concentrated within the **~1500–1700 cm⁻¹** protein-associated region.

CONCLUSION

These findings demonstrate the strong potential of combining FTIR spectroscopy with explainable machine learning for spectral phenotyping and biomarker-oriented cancer research.
Independent external validation and prospective clinical studies are required before diagnostic application.

Key Performance (Test Set)

Accuracy (XGBoost): **97.50%**

Best ROC-AUC (Random Forest): **0.9983**

Class AUC (XGBoost): **0.9961–1.0000**

Key Spectral Region: **~1500–1700 cm⁻¹**

1. BIOLOGICAL RATIONALE

EGFR
Epidermal Growth Factor Receptor
A key driver of cancer cell proliferation, survival and treatment response.

ALK
Anaplastic Lymphoma Kinase
A driver of oncogenesis and a promising therapeutic target in cancer.

ROS1
c-ROS Oncogene 1 Receptor Tyrosine Kinase
A druggable actionable driver in cancer.

2. FTIR SPECTROSCOPY

COLOR CODE	WAVENUMBER RANGE (cm ⁻¹)	VIBRATION ASSIGNMENT	MAJOR BIOMOLECULAR CONTRIBUTORS	TYPICAL INFORMATION PROVIDED
~3300–3400 cm ⁻¹	O–H, N–H stretching	Proteins, carbohydrates	Hydrogen bonding, hydration level, protein secondary structure, polysaccharides.	
~2920–2850 cm ⁻¹	C–H stretching	Lipids	Aliphatic chains of lipids, membrane composition, lipid content.	
~1740 cm ⁻¹	C=O stretching	Lipids, esters	Carbonyl groups of esters and lipids.	
~1650 cm ⁻¹	Amide I	Proteins (C–O stretching)	Protein secondary structure (α-helix, β-sheet), peptide backbone.	
~1540 cm ⁻¹	Amide II	Proteins (N–H bending)	Protein structure, hydrogen bonding and conformation.	
~1230–1080 cm ⁻¹	Phosphate / C–O stretching	Nucleic acids, carbohydrates	Phosphate backbone of DNA/RNA, C–O bonds in carbohydrates.	
~900–1000 cm ⁻¹	C–C, C–O vibrations	Biochemical fingerprint (region-specific)	Unique "fingerprint" region for tissue identification and comparison.	

Note: The exact peak positions may vary slightly depending on the sample type, biochemical composition and experimental conditions.

3. ANALYTICAL WORKFLOW

FTIR Spectroscopy + Machine Learning Pipeline

From Raw Spectra to Interpretable Biomarker Classification

- FTIR Spectra
- Preprocessing: SNV, Smoothing, Normalization
- Dimensionality Reduction: Principal Component Analysis (PCA)
- Machine Learning: Logistic Regression, Random Forest, XGBoost
- Model Evaluation: Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix
- Model Interpretation: SHAP - Important Wavenumbers, Biological Regions

OUTPUT: Cancer-Associated Spectral Discrimination

Accurate classification of FTIR spectra for biomarker-driven cancer profiling

4. DATASET & METHODOLOGY

DATASET OVERVIEW

1,000 samples in total; 250 samples per class: Healthy, EGFR, ALK and ROS1.
20 FTIR-derived features: 10 peak positions (wavenumbers) + 10 intensities.
80/20 stratified train-test split: 800 samples for training; 200 for testing.
Standard Normal Variate (SNV) preprocessing, mean centering and scaling.
PCA for dimensionality reduction and data visualization.
Random Forest and XGBoost for multi-class classification.
Evaluation metrics: accuracy, precision, recall, F1-score and ROC-AUC.

CLASS DISTRIBUTION

Healthy: 250 (25%)
EGFR: 250 (25%)
ALK: 250 (25%)
ROS1: 250 (25%)

FTIR-DERIVED FEATURES (20)

10 PEAK POSITIONS (cm⁻¹): 950, 1030, 1080, 1240, 1330, 1450, 1530, 1580, 1620, 1700
10 PEAK INTENSITIES: Intensities at the corresponding 10 peak positions

PREPROCESSING

Raw Spectra → SNV → Smoothing → Normalization → Mean Centering & Scaling

MODELING

Random Forest, XGBoost

EVALUATION

Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

METHODOLOGY WORKFLOW

1. FTIR Spectra → 2. Preprocessing → 3. PCA → 4. Modeling (Random Forest & XGBoost) → 5. Evaluation (Performance metrics & confusion matrix) → 6. Interpretation (SHAP analysis & key features) → 7. Reliable classification of FTIR spectra for cancer biomarker analysis

5. RESULTS

5.1 Model Performance Comparison

A. F1-score Comparison

Macro-Average F1-score: Logistic Regression (0.9450), Random Forest (0.9700), XGBoost (0.9750)

B. ROC-AUC Comparison

Macro-Average ROC-AUC: Logistic Regression (0.9923), Random Forest (0.9983), XGBoost (0.9981)

C. Corrected Macro-Average ROC Curve

Logistic Regression (AUC = 0.9939), Random Forest (AUC = 0.9987), XGBoost (AUC = 0.9985)

D. Final Performance Comparison (All Metrics)

Final Performance Comparison of FTIR Classification Models: Accuracy, Precision, Recall, F1-score, ROC-AUC

5.2 Spectral Analysis & Biomarker Interpretation

A. Mean FTIR Spectra by Class

Healthy, EGFR, ALK, ROS1

B. Top 10 Consensus Biomarkers (SHAP Importance)

Wavenumbers (cm⁻¹): 1700, 1640, 1510, 1520, 1620, 1580, 1530, 1590, 1610, 1630. Importance: 15.2, 12.9, 10.5, 7.6, 7.1, 5.7, 5.4, 3.7, 3.2, 2.7.

5.3 Classification Performance: Confusion Matrices (Test Set)

A. Logistic Regression

	Healthy	EGFR	ALK	ROS1
Healthy	240	4	2	4
EGFR	6	234	5	5
ALK	6	7	231	6
ROS1	4	5	6	245

Accuracy: 0.9450

B. Random Forest

	Healthy	EGFR	ALK	ROS1
Healthy	245	2	1	2
EGFR	2	246	1	1
ALK	1	2	245	2
ROS1	2	1	2	245

Accuracy: 0.9700

C. XGBoost

Accuracy: 0.9750

5.4 Key Findings

- XGBoost achieved the best overall performance: F1-score = 0.9750; ROC-AUC = 0.9981.
- Random Forest achieved the highest ROC-AUC: 0.9983.
- All models demonstrate excellent discrimination ability (ROC-AUC > 0.99).
- Spectral differences among classes are most pronounced in the ~1500–1700 cm⁻¹ protein-associated region.
- SHAP analysis identifies consensus biomarkers that drive class discrimination.

5. RESULTS — TABLES

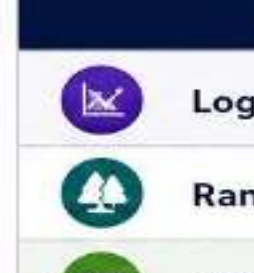
Table 1.	Comparative performance on the test set					
	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
	Logistic Regression	0.9450	0.9451	0.9450	0.9450	0.9923
	Random Forest	0.9700	0.9700	0.9700	0.9700	0.9983
	XGBoost	0.9750	0.9751	0.9750	0.9750	0.9981

Table 2.	Five-fold cross-validation	
	Model	Accuracy (mean ± SD)
	Logistic Regression	0.9520 ± 0.0117
	Random Forest	0.9800 ± 0.0032
	XGBoost	0.9750 ± 0.0063

TEST-SET HIGHLIGHT

- XGBoost achieved the highest accuracy, precision, recall and F1-score (0.9750).
- Random Forest achieved the highest ROC-AUC (0.9983).

CROSS-VALIDATION HIGHLIGHT

- Random Forest achieved the highest mean accuracy (98.00%).
- Lowest variability with SD = 0.0032.

Values correspond to the corrected FTIR classification analysis; bold values indicate the best value for the corresponding metric.

6. CONCLUSION & PERSPECTIVES

6.1 CONCLUSION

FTIR spectroscopy captures reproducible biochemical fingerprints of serum samples from four clinical groups (Healthy, EGFR, ALK, ROS1), with distinct variations particularly in the protein-associated region (~1500–1700 cm⁻¹) and the fingerprint region (~1200–1000 cm⁻¹).
After rigorous preprocessing, dimensionality reduction (PCA) and feature selection, FTIR-derived features enabled effective discrimination among the four classes.
All machine learning models demonstrated excellent performance, with ROC-AUC > 0.99 for all models.
XGBoost achieved the best overall performance: F1-score = 0.9750, Accuracy = 0.9750, ROC-AUC = 0.9981.
Random Forest achieved the highest ROC-AUC (0.9983) and the most stable performance in cross-validation (Accuracy = 0.9800 ± 0.0032).
SHAP analysis identified consensus discriminative wavenumbers, primarily in the ~1500–1700 cm⁻¹ region, which are consistent with known biochemical markers related to protein and lipid alterations.
These results demonstrate the potential of combining FTIR spectroscopy with machine learning for accurate, interpretable, and non-invasive biomarker-based classification.

6.2 PERSPECTIVES

Larger and multicentric studies

- Validate the models on larger, independent cohorts and across multiple centers to confirm generalizability.
- Include additional classes (e.g., other oncogenic drivers) and early-stage samples.

Integration with other data modalities

- Combine FTIR with clinical, genomic, and proteomic data to develop multimodal predictive models.
- Integrate liquid biopsy analytes (ctDNA, CTCs, exosomes) to improve early detection and monitoring.

Model optimization and interpretability

- Explore advanced deep learning and ensemble approaches.
- Further refine SHAP-based biomarkers for mechanistic interpretation and biological validation.

Clinical translation

- Develop a robust, rapid, and cost-effective FTIR-based diagnostic tool for clinical laboratories.
- Conduct prospective studies to assess the impact on patient management and outcomes.

Towards personalized medicine

- Use the model for patient stratification and targeted therapy selection.
- Monitor treatment response and disease progression through non-invasive spectral profiling.

This study provides strong evidence that FTIR spectroscopy combined with machine learning is a powerful approach for accurate, interpretable classification of biomarker-driven cancer subtypes. With further validation and clinical integration, it holds promise for advancing non-invasive precision oncology.